

Week 7 Mini Project

E-Commerce Sales Data — Exploration Worksheet

Name: _____

Date: _____

Dataset: E-Commerce Sales

This worksheet will guide you through a full data analysis of a real e-commerce sales dataset. Work through each section in order. You will clean the data, explore it, create charts, write down what you find, and then give real business recommendations based on your findings. There is no single right answer — your job is to think like a data analyst.

SECTION 1 — Load and First Look

Before you clean or analyse anything, you need to understand what you are working with. Load the dataset and answer the questions below.

Q1. How many rows and columns does the dataset have?

Hint: Use `.shape` on your `DataFrame`.

Your answer:

Q2. List all the column names. Which columns do you think will be most useful for analysis and why?

Hint: Use `.columns` or `.info()` to see the column names.

Your answer:

Q3. What does one row in this dataset represent? (What is one record?)

Your answer:

Q4. Which columns contain numbers, and which ones contain text or categories?

Hint: Look at the dtype shown in .info().

Your answer:

SECTION 2 — Data Cleaning

Real data is almost never perfect. Before you can trust your analysis, you need to find and fix problems in the data.

Q5. How many missing values are there in each column? Which column has the most missing values?

Hint: Use .isnull().sum() to count missing values per column.

Your answer:

Q6. How did you decide to handle the missing values — did you drop the rows, fill them in, or leave them? Explain your decision.

Your answer:

Q7. Did you find any duplicate rows? How many, and what did you do with them?

Hint: Use .duplicated().sum() to check.

Your answer:

Q8. Are there any column names that are hard to read or have spaces? How did you clean them?

Hint: You can use .str.lower() and .str.replace() on df.columns.

Your answer:

■ A good rule: after cleaning, run `.info()` and `.isnull().sum()` one more time to confirm everything looks right before moving on.

SECTION 3 — Exploration and Groupby

Now it is time to dig in. Use `groupby`, `value_counts`, and `describe` to start finding patterns.

Q9. Which country placed the most orders? What percentage of total orders does that country represent?

Hint: Try `.value_counts()` on the country column, then divide by the total number of rows.

Your answer:

Q10. What are the top 5 best-selling products? How did you decide what 'best-selling' means — by number of orders or by total quantity sold?

Hint: You can use `groupby` with `.sum()` on the quantity column, then sort.

Your answer:

Q11. Which month had the highest total sales revenue? Which month had the lowest?

Hint: You may need to extract the month from a date column first using `.dt.month`.

Your answer:

Q12. What is the average order value (the average amount spent per order)? What is the highest single order value in the dataset?

Hint: Use `.describe()` or calculate revenue per row first ($\text{quantity} \times \text{unit price}$).

Your answer:

Q13. Are there any products that appear to be returned or cancelled? How can you tell from the data?

Hint: Look for negative quantities or a status/cancellation column.

Your answer:

SECTION 4 — Visualisation

Create at least 5 charts. For each chart, write what you see. Do not just describe the chart — tell us what it means.

Chart 1 — Bar Chart

Create a bar chart showing the top 10 countries by number of orders.

After creating the chart, answer this: What does this chart tell you about where this business gets most of its customers?

Your answer:

Chart 2 — Line Chart

Create a line chart showing total revenue or orders by month.

After creating the chart, answer this: Is there a clear trend or seasonal pattern? When does the business seem busiest?

Your answer:

Chart 3 — Histogram

Create a histogram of the order values (revenue per order).

After creating the chart, answer this: Are most orders small or large? Is the distribution skewed? What does that tell you about the typical customer?

Your answer:

Chart 4 — Bar or Pie Chart

Create a chart showing the split between your top product categories or customer types.

After creating the chart, answer this: Which group is most important to this business?

Your answer:

Chart 5 — Seaborn Chart (your choice)

Use Seaborn to create one chart of your choice — for example a boxplot of revenue by country, or a heatmap of monthly sales by product.

After creating the chart, answer this: Why did you choose this chart? What question were you trying to answer?

Your answer:

SECTION 5 — Insights

Write 5 clear findings from your analysis. Each insight should connect to one of your charts or groupby results. Write in plain language — imagine you are telling a manager what you found.

Insight 1

Your finding:

Insight 2

Your finding:

Insight 3

Your finding:

Insight 4

Your finding:

Insight 5

Your finding:

■ A strong insight is specific. Instead of writing 'sales were high in December', try 'December had 3x more revenue than the lowest month, January, suggesting a strong seasonal peak.'

SECTION 6 — Business Recommendations

This is where your analysis becomes valuable. Based on everything you found, answer the questions below and write at least 3 recommendations for the business. Each recommendation must be linked to a specific finding from your data.

Q14. Looking at your top countries by orders — should the business focus more resources on its biggest market, or try to grow in the smaller markets? What does your data suggest?

Your answer:

Q15. Based on your monthly sales trend, when should the business run its biggest promotions? When should it prepare its stock and staffing levels?

Your answer:

Q16. Looking at the best-selling and worst-selling products — what would you recommend the business does with the products at the bottom of the list?

Your answer:

Q17. If the business asked you: 'What one thing should we do differently to increase revenue?' — what would you say, and what data supports that answer?

Your answer:

Your 3 Final Recommendations

Write your top 3 recommendations below. Each one should start with an action word (Focus, Reduce, Increase, Investigate, Launch, etc.) and be linked to your data.

Recommendation 1:

Recommendation 2:

Recommendation 3:

REFLECTION PARAGRAPH

Write a short paragraph (4 to 6 sentences) about what this dataset revealed to you. What surprised you? What was harder than you expected? What would you investigate next if you had more time?
